MATHEMATICS

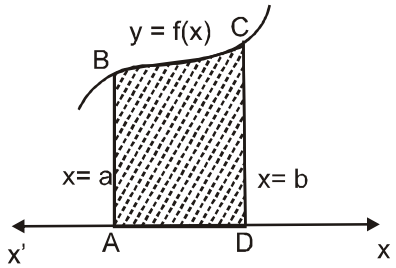
AREA UNDER CURVE
THEORY SHEET
ENGLISH

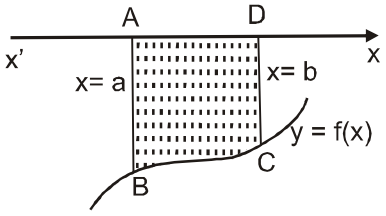
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1.1 AREA BOUNDED BY A CURVE

Area ABCDA bounded by the curve y = f(x), x-axis and two ordinates x = a and x = b is given by

$$\int_a^b |y|\,dx = \begin{cases} \int_a^b y\,dx, & \text{if } y \geq 0 \text{ for } x \in [a, b] \\ -\int_a^b y\,dx, & \text{if } y \leq 0 \text{ for } x \in [a, b] \end{cases}$$



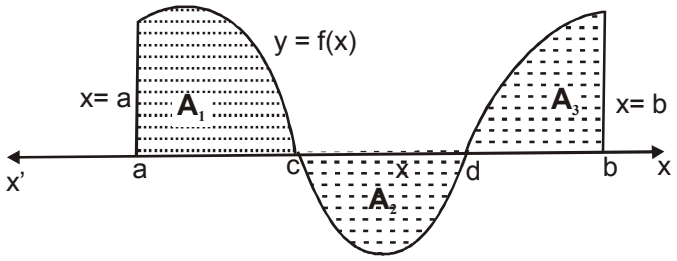


If however y i.e., f(x) changes sign in interval [a, b], say $y \geq 0$ in [a, c], $y \leq 0$ in [c, d] and $y \geq 0$ in [d, b], where a < c < d < b, then

area bounded by the curve y = f(x), x-axis and the lines x = a and x = b

$$= \int_a^b |y|\,dx = \int_a^c y\,dx - \int_c^d y\,dx + \int_d^b y\,dx$$

$= A_1 - A_2 + A_3$, where A_1, A_2 and A_3 are algebraic areas.

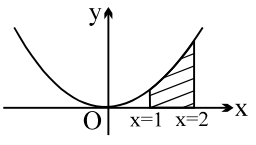


Questions

1. Find area bounded by curve $y = x^2$ x - axis and lines x = 1 and x =2

Ans 7/3

Sol $A = \int_1^2 x^2\,dx = 7/3$



2. Find the area bounded by the curve y = cos x between x = 0 and x = 2π.

Ans. 4

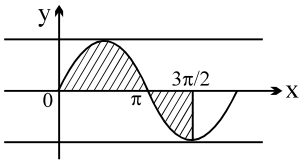
Sol. $A = \int_0^{2\pi} |\cos x|\,dx = \int_0^{\frac{\pi}{2}} \cos x\,dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -\cos x\,dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x\,dx = 4$

3. Find area enclosed between y = sin x & x-axis as x varies from 0 to $\frac{3\pi}{2}$.

Ans 3

Sol. $A = \int_0^{\pi} \sin x\,dx + \int_{\pi}^{3\pi/2} |\sin x|\,dx$

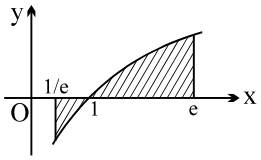
$2 + 1 = 3$



4. Find area bounded by curve y = *l*n x, x-axis and the two ordinate at x = $\frac{1}{e}$ and x = e.

Ans $2 - \frac{2}{e}$

Sol. $A = \int_{1/e}^{e} |\ln x|\,dx = 2 - \frac{2}{e}$



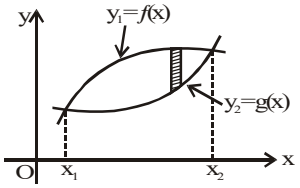
1.2 AREA BOUNDED BETWEEN TWO CURVES

(a) Area bounded by two curves y = *f* (x) & y = g(x)

such that *f* (x) > g(x) is

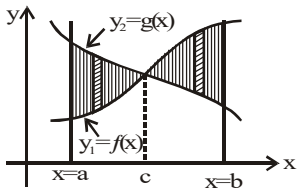
$A = \int_{x_1}^{x_2} (y_1 - y_2)\,dy$

$A = \int_{x_1}^{x_2} [f(x) - g(x)]\,dx$



(c) If the curves $y_1 = f$ (x) and y_2 = g(x) intersect at x = c, then required area

$A = \int_a^c (g(x) - f(x))\,dx + \int_c^b (f(x) - g(x))\,dx$

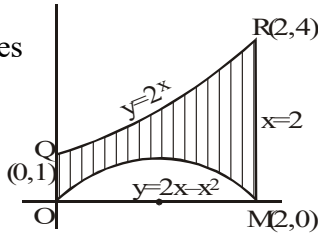


In both cases A = $= \int_a^b |f(x) - g(x)|\,dx$

Q.1 Compute the area of the figure bounded by the straight lines x = 0, x = 2 and the curves $y = 2^x$, $y = 2x - x^2$.

Ans. $\frac{3}{\ln 2} - \frac{4}{3}$

Sol. Figure is self-explanatory $y = 2^x$, $(x-1)^2 = -(y-1)$

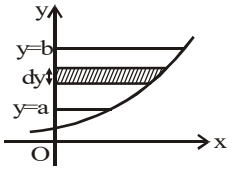


The required area $= \int_0^2 (y_1 - y_2)\,dx$

where $y_1 = 2^x$ and $y_2 = 2x - x^2 = \int_0^2 (2^x - 2x + x^2)dx$

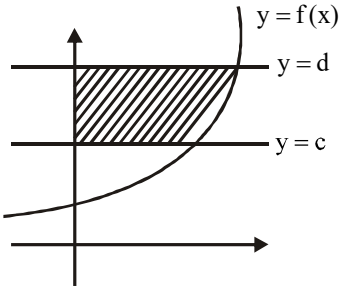
$= \left[\frac{2^x}{\ln 2} - x^2 + \frac{1}{3}x^3\right]_0^2 = \left(\frac{4}{\ln 2} - 4 + \frac{8}{3}\right) - \frac{1}{\ln 2} = \frac{3}{\ln 2} - \frac{4}{3}$ sq.units.

1.3 INTEGRATING WITH RESPECT TO Y-AXIS



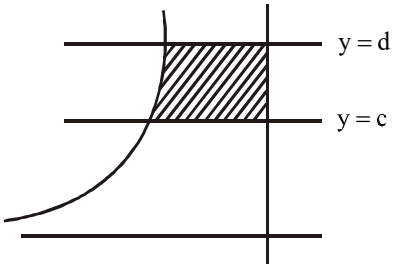
Sometimes integration w.r.t. y is very useful (horizontal strip) :

1. Area bounded by the curve, y-axis and the two abscissae at $y = a$ & $y = b$ is written as $A = \int_a^b x\,dy$.

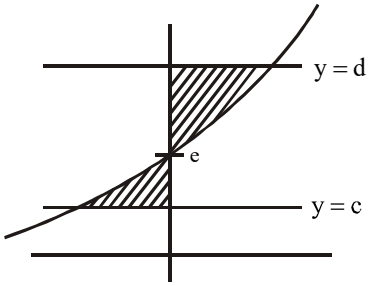


Ex:1. Area $= \int_c^d (x_f).dy$

$x_f \equiv x$ in terms of y



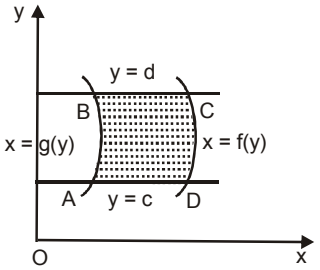
Ex:2. Area $= -\int_c^d x_f.dy$



Ex:3. Area $= -\int_c^e x_f.dy + \int_e^d x_f.dy$

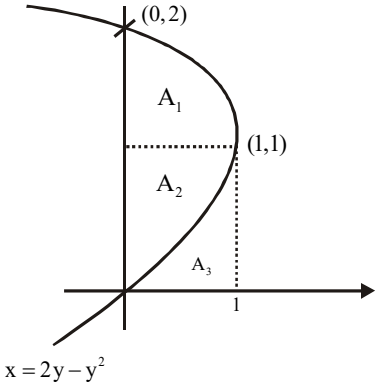
2. Area bounded by the two curves $x = f(y)$, $x = g(y)$ and two abscissae $y = c$ and $y = a$ is given by area ABCDA

$$= \int_c^d |x_1 - x_2|\,dy = \begin{cases} \int_c^d (x_1 - x_2)dy, \text{ if } x_1 \geq x_2 \text{ for } c \leq y \leq d \\ -\int_c^d (x_1 - x_2)dy, \text{ if } x_1 \leq x_2 \text{ for } c \leq y \leq d \end{cases}$$



Q.1 Curve $x = 2y - y^2$ and the y-axis

Ans. $\frac{1}{3}$



$y^2 - 2y + x = 0 \Rightarrow \quad y = \frac{2 \pm \sqrt{4-4x}}{2} = 1 \pm \sqrt{1-x}$

$y_1 = 1 + \sqrt{1-x}$; $\quad y_2 = 1 - \sqrt{1-x}$

1. With respect to x-axis

$A_2 = \int_0^1 (1 - y_2).dx = \int_0^1 (\sqrt{1-x}).dx$; $\quad A_1 = \int_0^1 (y_1 - 1).dx = \int_0^1 (\sqrt{1-x}).dx$; $A_3 = \int_0^1 y_1.dx = \int_0^1 \left(1 - \sqrt{1-x}\right).dx$

2. w.r.t y-axis

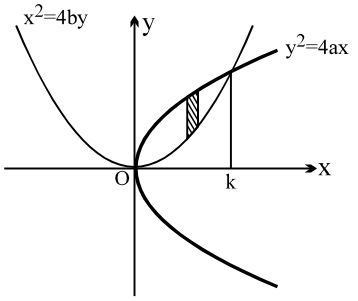
$A_1 = \int_1^2 (2y - y^2).dy$; $\quad A_2 = \int_0^1 (2y - y^2).dy$; $\quad A_3 = \int_0^1 (1 - (2y - y^2)).dy$

1.4 IMPORTANT RESULTS

1. Area enclosed between the parabolas $y^2 = 4ax$ & $x^2 = 4by$ is $\frac{16ab}{3}$ sq.units.

$y^2 = 4ax$; $x^2 = 4by$

$a > 0$; $b > 0$



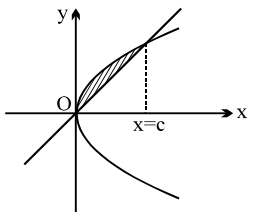
$$A = \int_0^k \left(2\sqrt{a}\sqrt{x} - \frac{x^2}{4b} \right) dx$$

$$A = \frac{16ab}{3}$$

2. Area included between the parabola $y^2 = 4ax$ & the line $y = mx$ is $\frac{8a^2}{3m^3}$ sq.units.

$y^2 = 4ax$ and $y = mx$

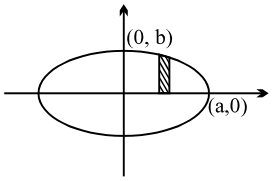
$$\int_0^c \left(2\sqrt{a}\sqrt{x} - mx\right)dx$$



$$A = \frac{8a^2}{3m^3}$$

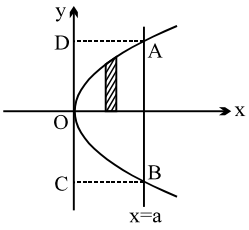
3. Whole area of the ellipse, $x^2/a^2 + y^2/b^2 = 1$ is π ab sq.units.

Whole area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



$$A = 4\int_0^a \left(b\sqrt{1 - \frac{x^2}{a^2}}\right)dx = \pi\ a\ b$$

4. Area enclosed by $y^2 = 4ax$ and its double ordinate at $x = a$ (chord perpendicular to the axis of symmetry)



$$2.\int_0^a \left(2\sqrt{ax}\right)dx$$

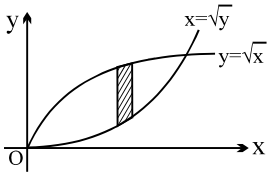
This simplifies to

Area of AOB = $\frac{2}{3}$ (area ABCD)

Q.1 Find area bounded by $y = \sqrt{x}$; $x = \sqrt{y}$

Ans $\frac{1}{3}$

Sol $y = \sqrt{x}$; $x = \sqrt{y}$



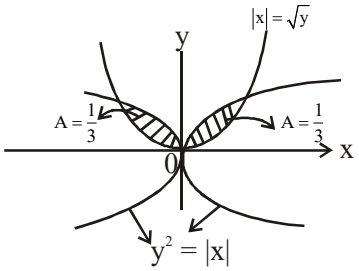
$$a = \frac{1}{4} \quad ; b = \frac{1}{4}$$

$$A = \frac{16}{3}$$

Q.2 Find area bounded by $y^2 = |x|$, $|x| = \sqrt{y}$

Ans. $\frac{2}{3}$

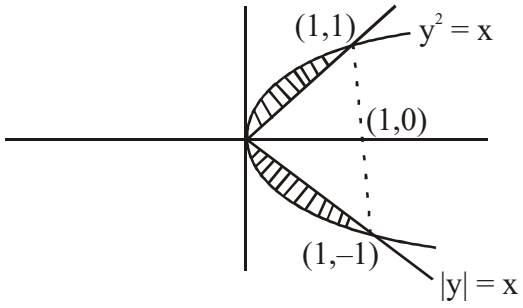
Sol.



Q. 3 Find area bounded by $y^2 = x$, $|y| = x$

Ans. $\frac{1}{3}$

Sol.



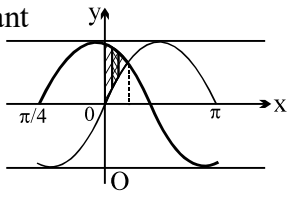
$$A = 2\int_0^1 \left(2\sqrt{\frac{1}{4}}\sqrt{x} - x\right) dx$$

$$= 2\left[\frac{2}{3}X^{\frac{3}{2}} - \frac{x^2}{2}\right]_0^1 = 2\left[\frac{2}{3} - \frac{1}{2}\right]$$

$$= \frac{1}{3}$$

1.5 QUESTIONS ON AREA BOUNDED

1. Find the area enclosed between y = sin x ; y = cos x and y-axis in the 1st quadrant



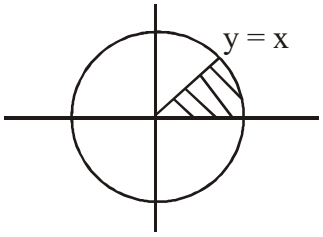
Ans. $\sqrt{2} - 1$

Sol. $A = \int_0^{\pi/4} (\cos x - \sin x)\, dx = \sqrt{2} - 1$

2. Find the area of the region in the first quadrant enclosed by the x-axis, the line y = x, and the circle $x^2 + y^2 = 32$.

Ans. 4π

Sol.



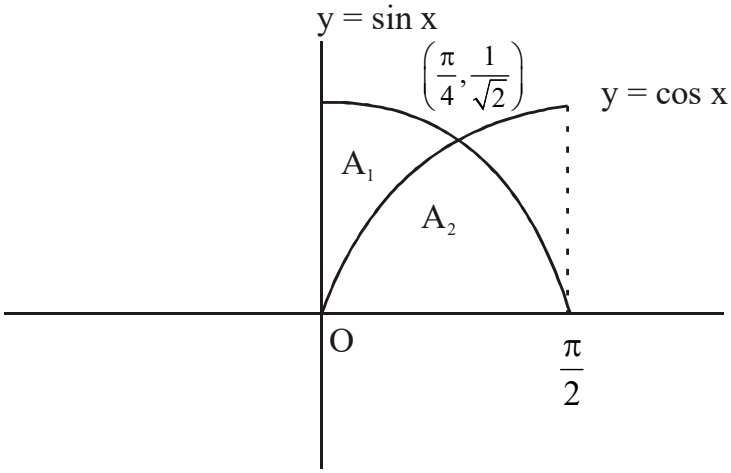
$$A = \frac{\pi r^2}{8}$$

$$= 4\pi$$

3. Find the ratio in which the area enclosed by the curve $y = \cos x$, $x \in \left[0, \frac{\pi}{2}\right]$ in the 1st quadrant is divided by the curve $y = \sin x$.

Ans. $1 : \sqrt{2}$

Sol.



$$A_1 + A_2 = \int_0^{\frac{\pi}{2}} \cos x dx = 1$$

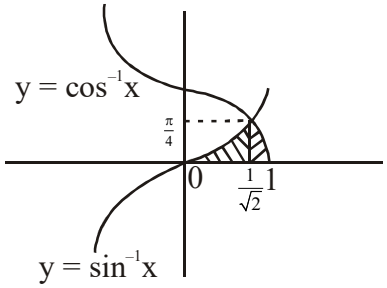
$$A_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\frac{\pi}{4}} = \sqrt{2} - 1$$

$$A_2 = 2 - \sqrt{2} \Rightarrow \frac{A_1}{A_2} = \frac{1}{\sqrt{2}}$$

4. Find the area enclosed between $y = \sin^{-1}x$; $y = \cos^{-1}x$ and the x-axis

Ans. $\sqrt{2} - 1$

Sol.



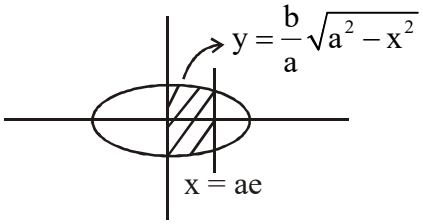
With respect to y-axis

$$A = \int_0^{\frac{\pi}{4}} (\cos y - \sin y)\,dy = (\sin y + \cos y)_0^{\frac{\pi}{4}} = \sqrt{2} - 1$$

5. Find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the ordinates x = 0 and x = ae, where, $b^2 = a^2 (1 - e^2)$ and e < 1.

Ans. $ab\left[e\sqrt{1-e^2} + \sin^{-1} e\right]$

Sol.



$$A = 2\int_0^{ae} \frac{b}{a}\sqrt{a^2 - x^2}\,dx$$

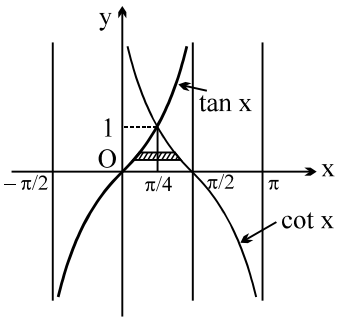
$$= \frac{2b}{a}\left[\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^{ae} = \frac{2b}{a}\left[\frac{ae}{2}\sqrt{a^2 - a^2e^2} + \frac{a^2}{2}\sin^{-1} e\right]$$

$$= ab\left[e\sqrt{1-e^2} + \sin^{-1} e\right]$$

6. Find Area enclosed by y = tan x ; y = cot x and x-axis in 1st quadrant.

Ans. *l*n(2)

Sol.

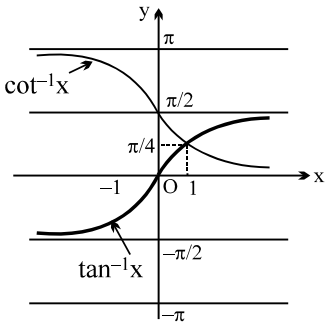


$$A = \int_0^{\pi/4} \tan x\,dx + \int_{\pi/4}^{\pi/2} \cot x\,dx$$

7. Find area bounded by $y = \tan^{-1} x$; $y = \cot^{-1} x$ and y-axis.

Ans. *l*n(2)

Sol.



$$A = \int_0^1 (\cot^{-1} x - \tan^{-1} x)\,dx$$

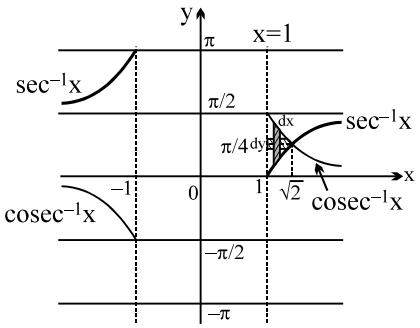
$$= \int_0^1 \left(\frac{\pi}{2} - 2\tan^{-1} x\right)dx$$

alternatively $A = 2\int_0^{\pi/4} x\,dy = 2\cdot\int_0^{\pi/4} \tan y\,dy$

8. Compute the area enclosed between $y = \sec^{-1}x$; $y = \text{cosec}^{-1}x$ and line $x - 1 = 0$

Ans. $\ln(3 + 2\sqrt{2}) - \dfrac{\pi}{2}$

Sol.



$$A = \int_1^{\sqrt{2}} (\text{cosec}^{-1}x - \sec^{-1} x)\,dx$$

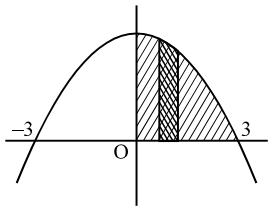
alternatively, if horizontal strip is used

$$A = 2\cdot\int_0^{\pi/4} (\sec y - 1)\,dy \qquad [\text{Ans. } \ln(3 + 2\sqrt{2}) - \frac{\pi}{2}\,]$$

9. Find area enclosed by $y = 9 - x^2$ and coordinates axes($x \geq 0$).

Ans. 18

Sol.

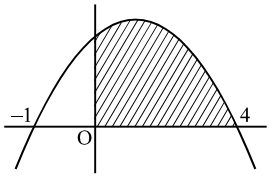


$$A = \int_0^3 (9 - x^2)\,dx \qquad [\text{Ans. 18 sq. units}]$$

10. Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes

Ans. 56/3

Sol.

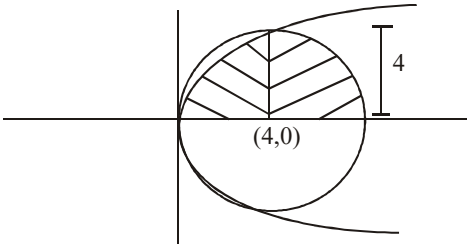


$$A = \int_0^4 y\,dx$$

11. Find the area lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside of the parabola $y^2 = 4x$.

Ans. $\frac{4}{3}(8+3\pi)$

Sol.



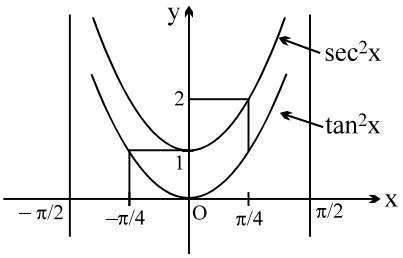
$$A = \frac{2ab}{3} + \frac{\pi r^2}{4}$$

$$= \frac{2\cdot4\cdot4}{3} + \frac{16\pi}{4} = \frac{32}{3} + 4\pi$$

12. Compute the area bounded by the curves $y = \sec^2 x$; $y = \tan^2 x$ and the line $x = -\frac{\pi}{4}$ and $x = \frac{\pi}{4}$

Ans. $\pi/2$

Sol $A = \int_{-\pi/4}^{\pi/4} (\sec^2 x - \tan^2 x)\,dx = 2\int_{0}^{\pi/4} (\sec^2 x - \tan^2 x)\,dx$

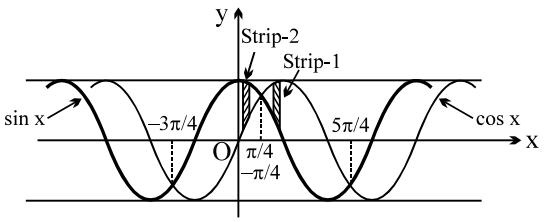


13. Curves $y = \sin x$; $y \cos x$ intersect each other at infinite number of points enclosing regions of equal areas. Compute the area of one such equal region.

Ans. $2\sqrt{2}$

Sol. (strip-1) $A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx$

(strip-2) $A = \int_{-3\pi/4}^{\pi/4} (\cos x - \sin x)\,dx$

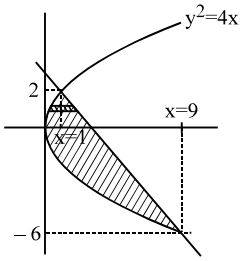


14. Find Area of the region bounded by the parabola $y^2 = 4x$ and a normal drawn to it with gradient – 1.

Ans. 64/3

Sol. Equation of normal

$y = mx - 2am - am^3$



$y = -x + 2 + 1$ ($a = 1$ and $m = -1$)

$y = 3 - x$; solving it will $y^2 = 4x$.

$(3 - x)^2 = 4x$

or $9 + x^2 - 6x = 4x$

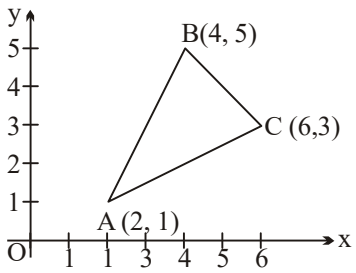
$x^2 - 10x + 9 = 0 \quad \Rightarrow \quad x = 1$ or $x = 9$

Now, $A = \int_{-6}^{2} x\,dy = \int_{-6}^{2}\left[(3-y) - \frac{y^2}{4}\right]dy$

15. Find area of triangle formed by (2,1), (6,3) and (4,5) with the help of integration (Only for Boards)

Ans. 6

Sol.



Equation of AB : $(y-1) = \left(\frac{5-1}{4-2}\right)(x-2) \Rightarrow y = 2x - 3$

Equation of BC : $(y-5) = \left(\frac{5-3}{4-6}\right)(x-4) \Rightarrow y = -x + 9$

Equation of CA : $(y-1) = \left(\frac{3-1}{6-2}\right)(x-2) \Rightarrow y = \frac{x}{2}$

Area of ΔABC = Area under AB + Area under BC – Area under AC

$= \int_{2}^{4}(2x-3)\,dx + \int_{4}^{6}(-x+9)\,dx - \int_{2}^{6}\frac{x}{2}\,dx$

$= 6 + 8 - 8$

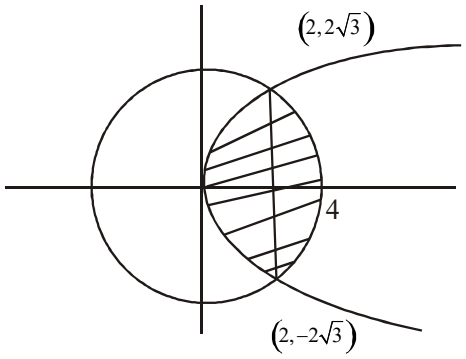
$= 6$

16. The area of the circle $x^2 + y^2 = 16$ exterior to the parabola $y^2 = 6x$ is

(A) $\frac{4}{3}\left(4\pi - \sqrt{3}\right)$ (B) $\frac{4}{3}\left(4\pi + \sqrt{3}\right)$ (C) $\frac{4}{3}\left(8\pi - \sqrt{3}\right)$ (D) $\frac{4}{3}\left(8\pi + \sqrt{3}\right)$

Ans. C

Sol.



$x^2 + y^2 = 16$

$y^2 = 6x$

Point of intersection $= x^2 + 6x - 16 = 0$

$x = 2$ or $x = -8$ (rejected)

$x = 2,\ y = \pm 2\sqrt{3}$

$$A = 2\left[\int_0^2 \sqrt{6x}\,dx + \int_2^4 \sqrt{16 - x^2}\,dx\right]$$

$$= 2\left[\frac{2}{3}\sqrt{6}\left(x^{\frac{3}{2}}\right)_0^2 + \left(\frac{x}{2}\sqrt{16 - x^2} + \frac{16}{2}\sin^{-1}\frac{x}{4}\right)_2^4\right]$$

$$= 2\left[\frac{8\sqrt{3}}{3} + \left(4\pi - 2\sqrt{3} - 8\sin^{-1}\frac{1}{2}\right)\right]$$

$$= \frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \qquad \text{(INTERIOR)}$$

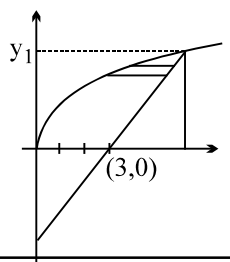
$$\text{EXTERIOR} = 16\pi - \frac{4\sqrt{3}}{3} - \frac{16\pi}{3}$$

$$= \frac{32\pi}{3} - \frac{4\sqrt{3}}{3} = \frac{4}{3}\left[8\pi - \sqrt{3}\right]$$

17. Find the area of the region in the 1st quadrant that is bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 3$.

Ans. $\dfrac{19 + 13\sqrt{13}}{12}$

Sol.



Point of intersection : $y = x - 3$ & $y = \sqrt{x} \Rightarrow x = y^2$

$y = y^2 - 3 \Rightarrow y^2 - y - 3 = 0 \Rightarrow y_1, y_2 = \frac{1 \pm \sqrt{13}}{2} \Rightarrow y_1 = \frac{\sqrt{13}+1}{2}$

$A = \int_0^{y_1} \left(y + 3 - y^2\right) dy = \frac{19 + 13\sqrt{13}}{12}$

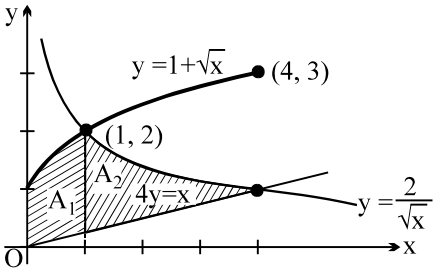
18. Find the area of the region in the 1st quadrant bounded on the left by the y-axis, below by the line $y = \frac{x}{4}$, above left by the curve $y = 1 + \sqrt{x}$ and above right by the curve $y = \frac{2}{\sqrt{x}}$

Ans. 11/3

Sol.(b) $A_1 = \int_0^1 \left((1 + \sqrt{x}) - \frac{x}{4}\right) dx = \frac{37}{24}$

$A_2 = \int_1^4 \left(\frac{2}{\sqrt{x}} - \frac{x}{4}\right) dx = \frac{17}{8}$

$A = A_1 + A_2 = \frac{11}{3}$



19. Find out the area enclosed by $y = x^2 + \cos x$ and its normal at $x = \frac{\pi}{2}$ in the first quadrant.

Ans. $\frac{\pi^5}{32} - \frac{\pi^4}{32} + \frac{\pi^3}{24} + 1$

Sol.

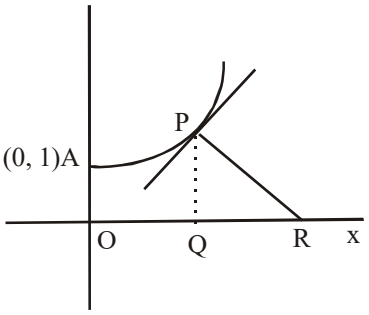
$f(x) = x^2 + \cos x \Rightarrow f'(x) = 2x - \sin x$

$\Rightarrow \quad f'\left(\frac{\pi}{2}\right) = \pi - 1$

Equation of normal at $x = \frac{\pi}{2}$ is

$$\left(y - \frac{\pi^2}{4}\right) = \frac{1}{1-\pi}\left(x - \frac{\pi}{2}\right)$$

at x axis, $y = 0 \Rightarrow x = \frac{(\pi-1)\pi^2}{4} + \frac{\pi}{2}$



Required area = area OAPQO + area of triangle PQR

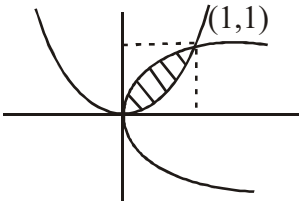
$$= \int_0^{\pi/2} (x^2 + \cos x)\, dx + \frac{1}{2}\left[\frac{(\pi-1)\pi^2}{4} + \frac{\pi}{2} - \frac{\pi}{2}\right]$$

$$= \left[\frac{x^3}{3} + \sin x\right]_0^{\pi/2} + \frac{(\pi-1)\pi^2}{8} = \frac{\pi^3}{24} + 1 + \frac{\pi^3}{8} - \frac{\pi^2}{8} = \left(\frac{\pi^3}{6} - \frac{\pi^2}{8} + 1\right) \text{ sq. units.}$$

2.1 AREA REPRESENTED BY INEQUALITIES

Q.1 $y^2 \le 4x \cap x^2 \le 4y$ or $\{(x, y) \mid y^2 \le 4x, x^2 \le 4y\}$

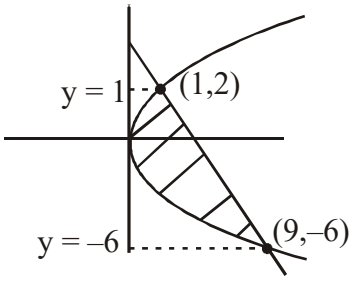
Sol.



$$A = \frac{16ab}{3} = \frac{16}{3}$$

Q.2 $y^2 \le 4x$, $x + y \le 3$

Sol.

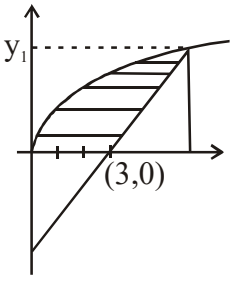


$$A = \int_{-6}^{1}\left(3 - y - \frac{y^2}{4}\right)dy$$

$$= \frac{64}{3}$$

Q.3 $y \le \sqrt{x}$, $y \ge 0, y \ge x - 3$

Sol.



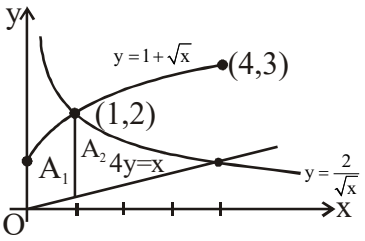
Point of intersection : $y = x - 3$ & $y = \sqrt{x} \Rightarrow x = y^2$

$$y = y^2 - 3 \Rightarrow y^2 - y - 3 = 0 \Rightarrow y_1 y_2 = \frac{1 \pm \sqrt{13}}{2} \Rightarrow y_1 = \frac{\sqrt{13} + 1}{2}$$

$$A = \int_0^{y_1} \left(y + 3 - y^2\right) dy = \frac{19 + 13\sqrt{13}}{12}$$

Q.4 $y \le 1 + \sqrt{x}$, $y \ge \frac{x}{4}$, $y \le \frac{2}{\sqrt{x}}$

Sol.



$$A_1 = \int_0^1 \left(\left(1 + \sqrt{x}\right) - \frac{x}{4}\right) dx = \frac{37}{24}$$

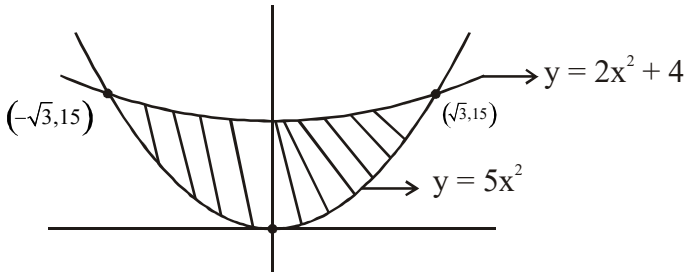
$$A_2 = \int_1^4 \left(\frac{2}{\sqrt{x}} - \frac{x}{4}\right) dx = \frac{17}{8}$$

$$A = A_1 + A_2 = \frac{11}{3}$$

Q. 5. The area of the region : $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is : **[JEE(Main)Feb.2021]**

Ans. $12\sqrt{3}$ square units

Sol.



$$\text{Required Area} = \int_{-\sqrt{3}}^{\sqrt{3}} (2x^2 + 9 - 5x^2)\, dx$$

$$= 12\sqrt{3}$$

2.2 SHIFTING OF ORIGIN

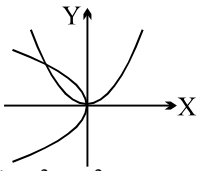
Since area remains invariant even if the co-ordinate axes are shifted, hence shifting of origin in many cases proves to be very convenient in computing the area.

Questions

1. Find area enclosed between the parabolas $y^2 - 2y + 4x + 5 = 0$ and $x^2 + 2x - y + 2 = 0$.

Ans. 4/3

[Hint:



(a) $(y-1)^2 = -4(x+1)$; $(x+1)^2 = y-1$

$Y^2 = -4X$ $\qquad$ $X^2 = Y$

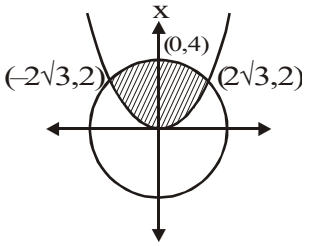
2. Find the area of the region common to the circle $x^2 + y^2 + 4x + 6y - 3 = 0$ and the parabola $x^2 + 4x = 6y + 14.$; $(y \geq -3)$

Ans. $\left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3}\right)$

Sol. Circle is $x^2 + y^2 + 4x + 6y - 3 = 0$

$\Rightarrow (x+2)^2 + (y+3)^2 = 16$

Shifting origin to $(-2,-3)$.



$X^2 + Y^2 = 16$

equation of parabola $\rightarrow (x+2)^2 = 6(y+3)$

$\Rightarrow X^2 = 6Y$

Solving circle & parabola, we get $X = \pm 2\sqrt{3}$

Hence they intersect at $\left(-2\sqrt{3},2\right)$ & $\left(2\sqrt{3},2\right)$

$$A = 2\left[\int_0^2 \sqrt{6Y}\,dY + \int_2^4 \sqrt{16 - Y^2}\,dY\right]$$

$$= 2\left[\frac{2}{3}\sqrt{6}\left[Y^{3/2}\right]_0^2 + \left[\frac{1}{2}Y\sqrt{16-Y^2} + \frac{16}{2}\sin^{-1}\frac{Y}{4}\right]_2^4\right] = \left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3}\right) \text{sq. units}$$

3. Find the area enclosed by the parabola $(y-2)^2 = x - 1$ and the tangent to it at (2, 3) and x-axis.

Ans. 9

Sol. Put $x - 1 = X$ and $y - 2 = Y$

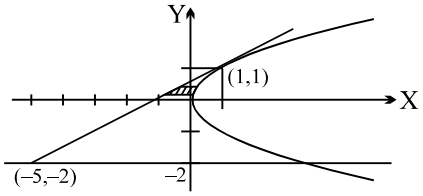
Hence the parabola becomes $Y^2 = X$

also $x = 2 \Rightarrow X = 1$

and $y = 2 \Rightarrow Y = 1$

also x-axis means $y = 0$

$\Rightarrow Y = -2$



Tangent : $YY_1 = 2 \cdot \frac{1}{4}(X + X_1)$

$2yy_1 = x + x_1$

$2y = x + 1;$ $\qquad$ Hence $A = \int_{-2}^{1} [Y^2 - (2Y - 1)]\,dY$

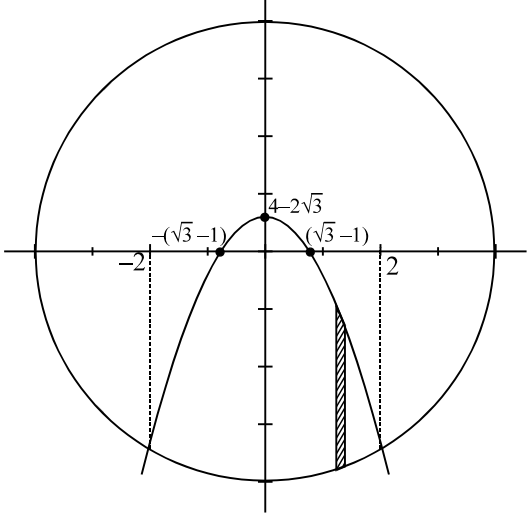
4. Area enclosed between the smaller arc of the circle $x^2 + y^2 - 2x + 4y - 11 = 0$ and the parabola $y = -x^2 + 2x + 1 - 2\sqrt{3}$

Ans. $\frac{32}{3} - 4\sqrt{3} + \frac{8\pi}{3}$

Sol. Circle : $(x-1)^2 + (y+2)^2 = 16$ $\qquad$ ....(1)

Parabola : $y = -[x^2 - 2x - 1 + 2\sqrt{3}]$

$= -[(x-1)^2 - 2 + 2\sqrt{3}$



$y + 2 = (4 - 2\sqrt{3}) - (x - 1)^2$ ....(2)

Let $x - 1 = X$ and $y + 2 = Y$

Hence Circle : $X^2 + y^2 = 16$;

Parabola : $Y = 4 - 2\sqrt{3} - X^2$

Solving the circle and parabola

$X = 2$ or -2

and $Y = -2\sqrt{3}$; $Y = 1 + 2\sqrt{3}$ (rejected)

$\therefore \quad A = 2\int_0^2 \left[(4 - 2\sqrt{3} - x^2) - \left\{ -\sqrt{16 - x^2} \right\} \right]$]

3. AREA WHEN SKETCHING OF CURVE IS SIGNIFICANT

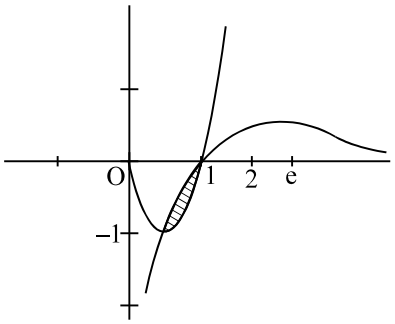
1. Area enclosed between the curves $y = ex \cdot l\text{n}\, x$ and $y = \dfrac{l\text{n}\, x}{ex}$

ans. $\dfrac{e}{4} - \dfrac{5}{4e}$

Sol. Solving $ex \cdot l\text{n}\, x = \dfrac{l\text{n}\, x}{ex}$

$l\text{n}\, x\,(e^2x^2 - 1) = 0$

$x = 1$ or $x = \dfrac{1}{e}$



where $x = 1$; $y_1 = 0$; $y_2 = 0$

$x = \dfrac{1}{e}$; $y_1 = -1$; $y_2 - 1$

Also examine the increasing and decreasing behaviour of the curve

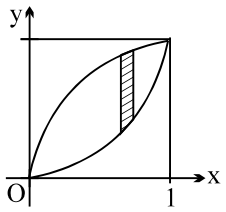
$$A = \int_{1/e}^{1} \left(\frac{l\text{n}\, x}{ex} - ex\, l\text{n}\, x \right) dx$$

2. Area enclosed by the curve $(y - \sin^{-1}x)^2 = x - x^2$.

ans. $\pi/4$

Sol. (a) $y - \sin^{-1}x = \pm \sqrt{x(1 - x)}$

$y = \sin^{-1}x + \sqrt{x(1-x)}$ or $y = \sin^{-1}x - \sqrt{x(1-x)}$

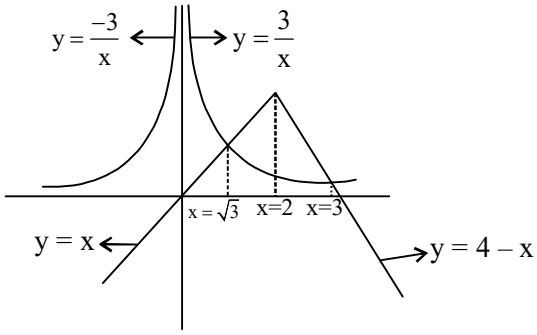


$$A = 2\int_0^1 \sqrt{x - x^2}\,dx$$

3. Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \dfrac{3}{|x|}$

Ans. $\dfrac{4 - 3\ell n 3}{2}$

[Sol. $y = \begin{cases} 2-(2-x) \text{ if } x \le 2 \\ \quad = x \\ 2+(2-x) \text{ if } x > 2 \\ \quad = 4 - x \end{cases}$

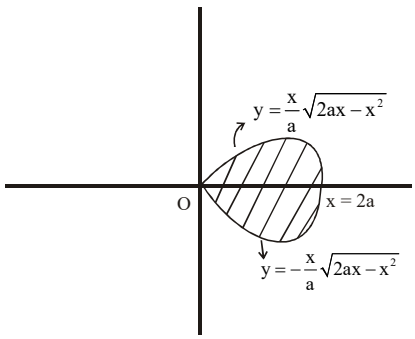


$$\text{Area} = \int_{\sqrt{3}}^{2} x\,dx + \int_2^3 (4-x)\,dx - \int_{\sqrt{3}}^{3} \frac{3}{x}\,dx = \frac{4 - 3\ell n 3}{2}$$

4. Find the whole area enclosed by the curve $a^2y^2 = x^3(2a - x)$, $a > 0$

ans. πa^2

Sol.



$$\text{Area} = 2\int_0^{2a} \frac{x}{a}\sqrt{2ax - x^2}\,dx$$

$$= \frac{2}{a}\left[\int_0^{2a} (x - a + a)\sqrt{2ax - x^2}\,dx\right]$$

$$= \frac{2}{a}\left[\int_0^{2a}(x-a)\sqrt{a^2-(x-a)^2}\,dx + a\int_0^{2a}\sqrt{a^2-(x-a)^2}\,dx\right]$$

(A) $\int_0^{2a}(x-a)\sqrt{a^2-(x-a)^2}\,dx$

Let $a^2-(x-a)^2 = t^2 \Rightarrow -(x-a)dx = tdt$

$$= \int_0^0 -t\cdot t\,dt = 0$$

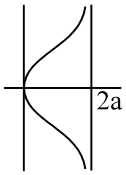
(B) $a\int_0^{2a}\sqrt{a^2-(x-a)^2}\,dx = a\left[\left(\frac{x-a}{2}\right)\sqrt{a^2-(x-a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x-a}{a}\right)\right]_0^{2a} = \frac{\pi a^3}{2}$

Area = (A) + (B) = $\frac{2}{a}\left[0+\frac{\pi a^3}{2}\right] = \pi a^2$

5. Find Area bounded by the curve $y^2(2a-x) = x^3$, $a > 0$ and its asymptote.

Ans. $3\pi a^2$

Sol.



$$A = 2\int_0^{2a} x\sqrt{\frac{x}{2a-x}}\,dx$$

Let $x = 2a\sin^2\theta \Rightarrow dx = 4a\sin\theta\cos\theta\,d\theta$

$$= 16a^2\int_0^{\frac{\pi}{2}}\sin^3\theta\cos\theta\tan\theta\,d\theta = 16a^2\int_0^{\frac{\pi}{2}}\sin^4\theta\,d\theta = 16a^2\times\frac{3\pi}{16}$$

$= 3\pi a^2$

6. Area of the loop $ay^2 = x^2(a-x)$, $a > 0$

ans. $\frac{8a^2}{15}$

Sol. $A = 2\int_0^a x\sqrt{\frac{a-x}{x}}\,dx$

Let $x = a\sin^2\theta \Rightarrow dx = 2a\sin\theta\cos\theta\,d\theta$

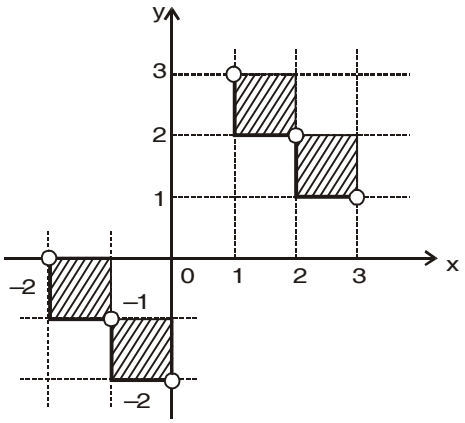
$$A = 2\int_0^{\frac{\pi}{2}} a\sin^2\theta(\cos\theta)2a\sin\theta\cos\theta\,d\theta$$

$$= 4a^2\int_0^{\frac{\pi}{2}}\sin^3\theta\cos^2\theta\,d\theta = \frac{8a^2}{15}$$

7. Find the area enclosed by the solution set of [x]. [y] = 2 Where [.] represent greatest integer function of x.

ans. 4

Sol [x]. [y] = 2



Here four cases arise

(1) $[x] = 2$ & $[y] = 1 \Rightarrow 2 \le x < 3$ & $1 \le y < 2$

(2) $[x] = 1$ & $[y] = 2 \Rightarrow 1 \le x < 2$ & $2 \le y < 3$

(3) $[x] = -2$& $[y] = -1 \Rightarrow -2 \le x < -1$ & $-1 \le y < 0$

(4) $[x] = -1$ & $[y] = -2 \Rightarrow -1 \le x < 0$ & $-2 \le y < -1$

8. Find the area of the region bounded by the curves $y = f(x)$, $y = |g(x)|$ and the lines $x = 0$, $x = 2$, where f and g are continuous functions satisfying $f(x + y) = f(x) + f(y) - 8xy$ for all $x, y \in R$

Ans. 4/3

***Solution*:**

Given, $f(x + y) = f(x) + f(y) - 8xy$ for all x, y

Putting $x = 0$ and $y = 0$, we get

$f(0) = f(0) + f(0) \Rightarrow f(0) = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) - f(x+0)}{h}$$

$$= \lim_{h \to 0} \frac{f(x) + f(h) - 8xh - f(x) - f(0) - 8x.0}{h}$$

$$= \lim_{h \to 0} \frac{f(h) - f(0)}{h} - 8x = f'(0) - 8x$$

Thus $f'(x) = 8 - 8x$ [$\because f'(0) = 8$]

Integrating both sides, we get

$f(x) = 8x - 4x^2 + c$ ... (2)

Putting $x = 0$, we get $f(0) = 0 + c \Rightarrow c = 0$ [$\because f(0) = 0$]

Hence $f(x) = 8x - 4x^2$ ... (3)

Given $g(x + y) = g(x) + g(y) + 3xy(x + y)$ for all x and y

Putting $x = y = 0$, we get $g(0) = 0$ ... (4)

Now $g'(x) = \lim_{h\to 0} \frac{g(x+h)-g(x)}{h} = \lim_{h\to 0} \frac{g(x+h)-g(x+0)}{h}$

$$= \lim_{h\to 0}\left[\frac{g(h)-g(0)}{h} + 3x^2 + 3xh\right]$$

$$= g'(0) + 3x^2 = -4 + 3x^2 \ [\because g'(0) = -4]$$

Thus $g'(x) = -4 + 3x^2$ ... (5)

$\therefore$ $g(x) = -4x + x^3 + k$

Putting $x = 0$, we get $g(0) = k$

$\Rightarrow$ $k = 0$ $[\because g(0)=0]$

$\therefore$ $g(x) = x^3 - 4x$ ... (6)

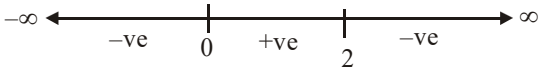
For points where $y = f(x)$ and $y = g(x)$ intersect,

$8x - 4x^2 = x^3 - 4x$

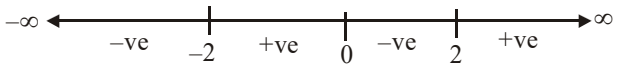
$\Rightarrow$ $x^3 + 4x^2 - 12x = 0$

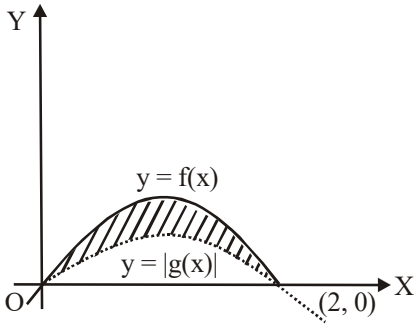
$\Rightarrow$ $x = 0, 2, -6$

Sign scheme for f(x) i.e., for $(8x - 4x^2)$ is



Sign scheme for g(x) i.e. for $x(x^2 - 4)$ is





$f(x) - |g(x)| = 8x - 4x^2 - (4x - x^3)$ $(\because\ g(x) \le 0 \text{ in } [0, 2])$

$= x^3 - 4x^2 + 4x = x(x-2)^2 \ge 0$

Area bounded by $y = f(x)$ and $y = |g(x)|$ between $x = 0$ and $x = 2$

$$= \int_0^2 (y_1 - y_2)\,dx = \int_0^2 [(8x - 4x^2) - (4x - x^3)]\,dx$$

$$= \int_0^2 (x^3 - 4x^2 + 4x)\,dx = \frac{4}{3} \text{ sq. units}.$$

and $g(x+y) = g(x) + g(y) + 3xy(x+y)$ for all $x, y \in R$. also $f'(0) = 8$ and $g'(0) = -4$

9. Find the area enclosed by the circle $x^2 + y^2 = 4$, the parabola $y = x^2 + x + 1$, the curve $y=\left[\sin^2\frac{x}{4}+\cos\frac{x}{4}\right]$ and the x-axis, (where [x] denotes the integral part of x).

Ans. $\left(\frac{2\pi}{3}+\sqrt{3}-\frac{1}{6}\right)$

Sol.

Equation of given circle is $x^2+y^2=4$ ... (1)

$\therefore \quad -2 \le x \le 2$ and $-2 \le y \le 2$

Let $z = \sin^2\frac{x}{4}+\cos\frac{x}{4} = 1-\cos^2\frac{x}{4}+\cos\frac{x}{4} = 1+t-t^2$, where $t=\cos\frac{x}{4}$

Now $-\frac{1}{2}\le\frac{x}{4}\le\frac{1}{2} \quad \therefore \quad 0<t\le 1 \Rightarrow 1\le z\le\frac{5}{4}$

for $x\in[2,2]$, curve $y=\left[\sin^2\frac{x}{4}+\cos\frac{x}{4}\right]$ becomes $y=1$ ... (2)

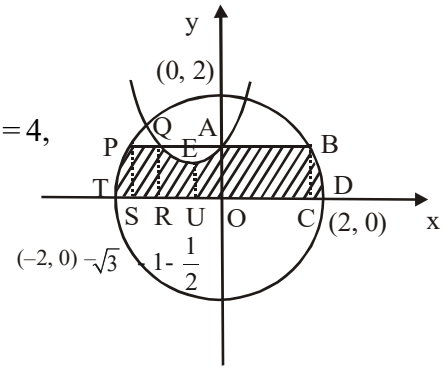
Given parabola is

$y = x^2 + x + 1$ ... (3)

or, $\left(x+\frac{1}{2}\right)^2 = y-\frac{3}{4}$ ... (4)

Its axis is $x=-\frac{1}{2}$ and vertex is $\left(-\frac{1}{2},\frac{3}{4}\right)$.

Now, we have to find out the area enclosed by the circle $x^2+y^2=4$,



parabola $\left(y-\frac{3}{4}\right)=\left(x+\frac{1}{2}\right)^2$, line $y = 1$ and x-axis

T	S	R	U	D
(–2, 0)	$(-\sqrt{3}, 0)$	(–1, 0)	(–1/2, 0)	(2, 0)

Required area is shown as shaded region in the figure.

Hence required area = area OABCO + area PQRS + area RQEAOR + 2 area CBDC.

$$\sqrt{3}\times 1+(\sqrt{3}-1)\times 1+\int_{-1}^{0}(x^2+x+1)dx+2\int_{\sqrt{3}}^{2}\left(\sqrt{4-x^2}\right)dx$$

$$=\left(2\sqrt{3}-1\right)+\left[\frac{x^3}{3}+\frac{x^2}{2}+x\right]_{-1}^{0}+2\left[\frac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\left(\frac{x}{2}\right)\right]_{\sqrt{3}}^{2}$$

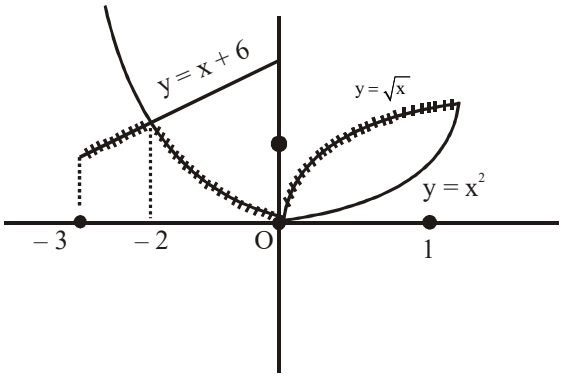
$$=\left(2\sqrt{3}-1\right)+\left[0-\left(-\frac{1}{3}+\frac{1}{2}-1\right)\right]+2\left[(0+\pi)-\left(\frac{\sqrt{3}}{2}+\frac{2\pi}{3}\right)\right]$$

$$=\left(2\sqrt{3}-1\right)+\frac{5}{6}+\frac{2\pi}{3}-\sqrt{3} \quad =\left(\frac{2\pi}{3}+\sqrt{3}-\frac{1}{6}\right)\text{sq. units}.$$

10. Let $f:[-3,1]\to R$ be given as $f(x)=\begin{cases}\min\left\{(x+6),x^2\right\} & ,-3\le x\le 0\\ \max\left\{\sqrt{x},\ x^2\right\} & ,0\le x\le 1\end{cases}$. If the area bounded by $y=f(x)$ and x-axis is A, then the value of 6A is equal to _____. **[JEE(Main)Mar.2021]**

Ans. 41

Sol.



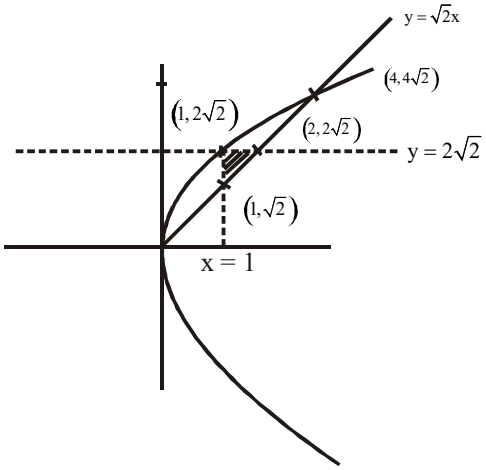
$$\int_{-3}^{-2}(x+6)\,dx+\int_{-2}^{0}x^2dx+\int_{0}^{1}\sqrt{x}\ dx$$

$$A=\frac{41}{6}$$

11. The area enclosed by $y^2=8x$ and $y=\sqrt{2}x$ that lies outside the triangle formed by $y=\sqrt{2}x$, $x=1$, $y=2\sqrt{2}$, is equal to :

Ans. $\frac{13\sqrt{2}}{6}$

Sol.



$$\text{Area of triangle} = \frac{1}{2}\times\sqrt{2}\times 1 = \frac{1}{\sqrt{2}}$$

$$\text{req. area} = \int_0^4 \left(2\sqrt{2}\sqrt{x} - \sqrt{2}x\right)dx - \frac{1}{\sqrt{2}}$$

$$= \left(2\sqrt{2}\,\frac{x^{3/2}}{\frac{3}{2}} - \sqrt{2}\,\frac{x^2}{2}\right)_0^4 - \frac{1}{\sqrt{2}}$$

$$= \frac{4\sqrt{2}(4)^{3/2}}{3} - \frac{16\sqrt{2}}{2} - \frac{1}{\sqrt{2}}$$

$$= \frac{32\sqrt{2}}{3} - \frac{16\sqrt{2}+\sqrt{2}}{2}$$

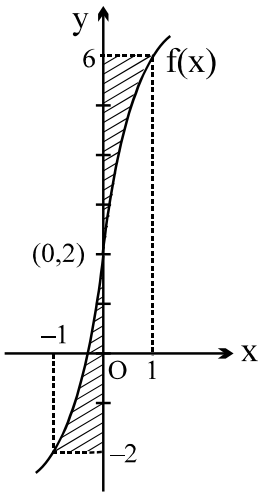
$$= \frac{64\sqrt{2}-51\sqrt{2}}{6} = \frac{13\sqrt{2}}{6}$$ Ans.

4. AREA BOUNDED BY INVERSE OF A FUNCTION

1. Let $f(x) = x^3 + 3x + 2$ and g (x) is the inverse of it. Find the area bounded by g (x), the x-axis and the ordinate at $x = -2$ and $x = 6$.

Ans. 9/2

[Sol. The required area will be equal to area enclosed by $y = f(x)$, the y-axis between the abscissa at $y = -2$ and $y = 6$



Hence $A = \int_0^1 (6 - f(x))dx + \int_{-1}^0 (f(x) - (-2))dx$

$= \int_0^1 (4 - x^3 - x)\,dx + \int_{-1}^0 (x^3 + 3x + 4)\,dx = 9/2$ Ans

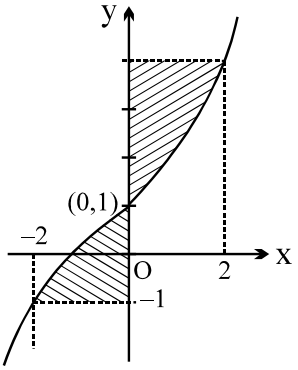
2. Find the area bounded by the curve g (x), the x-axis and the ordinate at x = – 1 and x = 4 where g (x) is the inverse of the function $f(x) = \frac{x^3}{24} + \frac{x^2}{8} + \frac{13x}{12} + 1$

Ans. $\frac{16}{3}$

Sol. f(x) is always monotonic with

f(0) = 1 ; f(2) = 4 and f(–2) = – 1

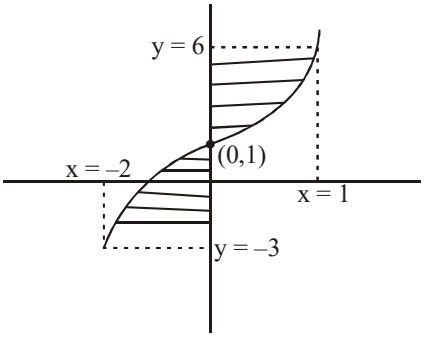
$A = \int_0^2 (4 - f(x))\,dx + \int_{-2}^0 (f(x) + 1)\,dx$]



3. f(x) = $x^3 + 2x^2 + 2x + 1$ and g (x) is the inverse of it. Then compute the area bounded by g (x), x-axis and the ordinate at x = – 3 and x = 6.

Ans. 101/12

Sol.



$f'(x) = 3x^2 + 4x + 2 \Rightarrow D < 0$

$f'(x) > 0 \ \forall \ x \in R$

$\Rightarrow f(0) = 1, f(-2) = -3, f(1) = 6$

$$A = \int_{-2}^{0} \left(f(x)+3\right)dx + \int_{0}^{1} \left(6 - f(x)\right)dx$$

$$= \left[\frac{x^4}{4} + \frac{2x^3}{3} + x^2 + 4x\right]_{-2}^{0} + \left[5x - \frac{x^4}{4} - \frac{2x^3}{3} - x^2\right]_{0}^{1}$$

$$= \left[0 - 4 + \frac{16}{3} - 4 + 8\right] + \left[5 - \frac{1}{4} - \frac{2}{3} - 1\right]$$

$$= \frac{16}{3} + \frac{37}{12}$$

$$= \frac{101}{12}$$

5.1 DETERMINATION OF PARAMETERS

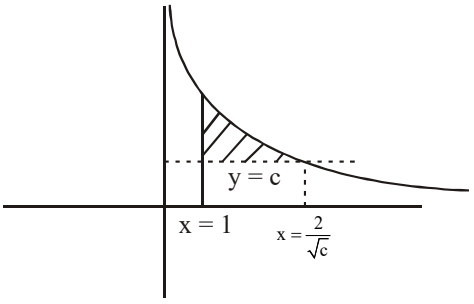
Ex 1. Find a if area bounded by $y = x^2 - 9$, x-axis, y - axis and x = a is (i) 16 (ii) 20

Questions

1. Find the value of c for which the area of the figure bounded by the curves $y = \frac{4}{x^2}$; x = 1 and y = c is equal to $\frac{9}{4}$.

Ans. $c = \frac{1}{4}$; $c = \frac{49}{4}$

Sol.



$$A = \int_{1}^{\frac{2}{\sqrt{c}}} \left(\frac{4}{x^2} - c\right)dx = \frac{9}{4}$$

$$\Rightarrow \left[-\frac{4}{x} - cx\right]_{1}^{\frac{2}{\sqrt{c}}} = \frac{9}{4}$$

$$\Rightarrow \left[-2\sqrt{c} - 2\sqrt{c} + 4 + c\right] = \frac{9}{4}$$

$\Rightarrow 4c - 16\sqrt{c} + 7 = 0$

$\Rightarrow 4c - 14\sqrt{c} - 2\sqrt{c} + 7 = 0$

$\Rightarrow \left(2\sqrt{c} - 7\right)\left(2\sqrt{c} - 1\right) = 0$

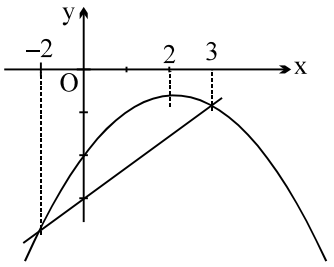
$\Rightarrow \sqrt{c} = \frac{7}{2}$ or $\frac{1}{2}$

$\Rightarrow c = \frac{49}{4}$ or $\frac{1}{4}$

2. Find the area of the figure bounded by the parabola $y = ax^2 + 12x - 14$ and the straight line $y = 9x - 32$ if the tangent drawn to the parabola at the point $x = 3$ is known to make an angle $\pi - \tan^{-1}6$ with the x-axis.

Ans. 125/2

Hint: $y = ax^2 + 12x - 14$



$$\frac{dy}{dx} = 2ax + 12 \quad ; \quad \left.\frac{dy}{dx}\right|_{x=3} = 6a + 12$$

hence $\tan(\pi - \tan^{-1}6) = 5a + 12$

$-6 = 6a + 12 \Rightarrow a = -3$

hence $y = -3x^2 + 12x - 14$ (note that $D < 0, y < 0 \ \forall \ x \in R$)

point of intersection of the line with parabola are $x = -2$ or 3

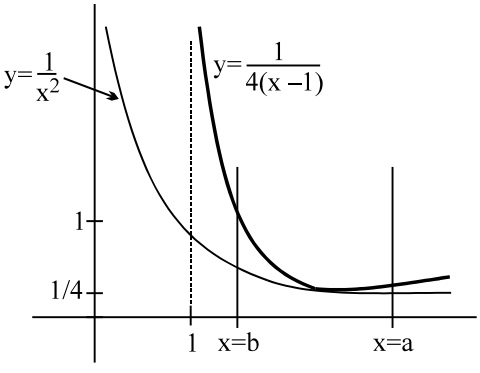
Hence $A = \int_{-2}^{3} [-3x^2 + 12x - 14] - (9x - 32)]dx$

3. Find the value of 'a' ($a > 2$) for which the reciprocal of the area enclosed between $y = \frac{1}{x^2}$; $y = \frac{1}{4(x-1)}$; $x = 2$ and $x = a$ is 'a' itself and for what values of $b \in (1, 2)$, the area of the figure bounded by the lines $x = b$ and $x = 2$ is $1 - \frac{1}{b}$.

Ans. $-e^2$

Sol. $x^2 = 4(x - 1)$

$(x - 2)^2 = 0 \Rightarrow$ curves touch other



$$\therefore \quad \int_{2}^{a}\left(\frac{1}{4(x-1)} - \frac{1}{x^2}\right)dx = \frac{1}{a}$$

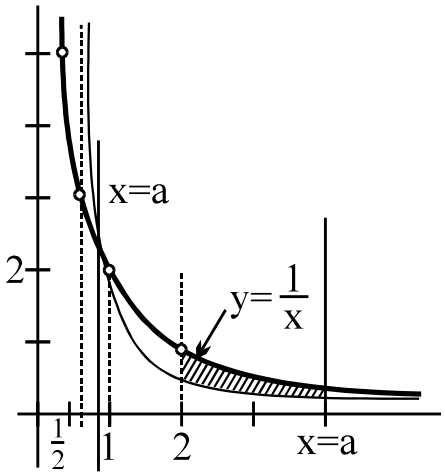
$a = e^2 + 1$ Ans.

Also $1 - \frac{1}{b} = \int_{b}^{2}\left(\frac{1}{4(x-1)} - \frac{1}{x^2}\right)dx \Rightarrow b = 1 + e^{-2}$

4. For what value of 'a' is the area of the figure bounded by the lines, $y=\frac{1}{x}$, $y=\frac{1}{2x-1}$, $x=2$ & $x=a$ $(a>2)$ equal to $\ln\frac{4}{\sqrt{5}}$?

Ans. $a=8, \frac{8}{15}$

Sol. $y=2x-1$; $y=\frac{1}{x}$; solving these two we get, $x=1$



Hence $A=\int_2^a\left(\frac{1}{x}-\frac{1}{2x-1}\right)dx=\ln\frac{4}{\sqrt{5}}$

$$\left[\ln x-\frac{1}{2}\ln(2x-1)\right]_2^a=\ln\frac{4}{\sqrt{5}}$$

$$\left[2\ln\frac{x^2}{2x-1}\right]_2^a=\ln\frac{16}{5}\Rightarrow\ln\left(\frac{a^2}{2a-1}\right)-\ln\frac{4}{3}=\ln\frac{16}{5}$$

$$\ln\frac{a^2}{2a-1}=\ln\frac{64}{15}\Rightarrow\frac{a^2}{2a-1}=\frac{64}{15}\Rightarrow 15a^2-128a+64=0$$

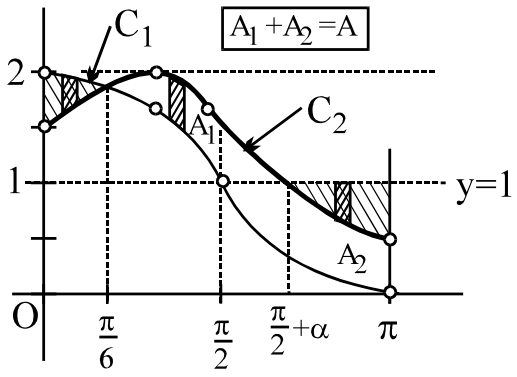
$a=8$; $a=\frac{8}{15}$

5. Consider the two curves $C_1: y=1+\cos x$ & $C_2: y=1+\cos(x-\alpha)$ for $\alpha\in\left(0,\frac{\pi}{2}\right)$; $x\in[0,\pi]$. Find the value of α, for which the area of the figure bounded by the curves C_1, C_2 & $x=0$ is same as that of the figure bounded by C_2, $y=1$ & $x=\pi$.

Ans. $\frac{\pi}{3}$

Sol. $1+\cos x=1+\cos(x-\alpha)$

$x=\alpha-x\quad\Rightarrow\quad x=\frac{\alpha}{2}$



now $\int_0^{\pi/2}(\cos x-\cos(x-\alpha))dx=-\int_{\frac{\pi}{2}+\alpha}^{\pi}(\cos(x-\alpha))dx$

or $[\sin x-\sin(x-\alpha)]_0^{\alpha/2}=[\sin(x-\alpha)]_{\pi}^{\frac{\pi}{2}+\alpha}$

$$\left[\sin\frac{\alpha}{2}-\sin\left(-\frac{\alpha}{2}\right)\right]-[0-\sin(-\alpha)] = \sin\left(\frac{\pi}{2}\right)-\sin(\pi-\alpha)$$

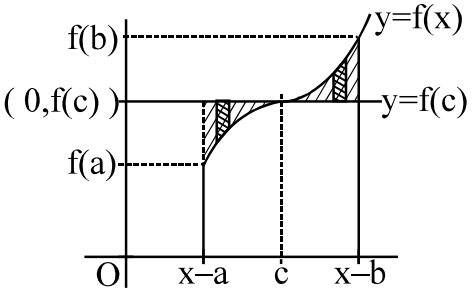
$$2\sin\frac{\alpha}{2} - \sin\alpha = 1 - \sin\alpha\,,$$

hence $\quad 2\sin\frac{\alpha}{2} = 1 \;\Rightarrow\; \alpha = \frac{\pi}{3}$

5.2 VARIABLE AREA GREATEST AND LEAST VALUE :

An important concept :

If $y = f(x)$ is a monotonic function in (a, b) then the area bounded by the ordinates at $x = a$, $x = b$, $y = f(x)$ and $y = f(c)$, [where $c \in (a, b)$] is minimum when $c = \frac{a+b}{2}$.



Proof : $A = \int_a^c (f(c) - f(x))dx + \int_c^b (f(x) - f(c))dx$

$$= f(c)\,(c-a) - \int_a^c (f(x))dx + \int_c^b (f(x))dx - f(c)\,(b-c)$$

$$= \{(c-a)-(b-c)\}\,f(c) + \int_c^b (f(x))dx - \int_a^c (f(x))dx$$

$$A = [2c-(a+b)]\,f(c) + \int_c^b (f(x))dx - \int_a^c (f(x))dx$$

differentiate w.r.t. c

$$\frac{dA}{dc} = [2c-(a+b)]\,f'(c) + 2f(c) + 0 - f(c) - (f(c) - 0)$$

for maxima and minima $\frac{dA}{dc} = 0$

$$f'(c)\,[2c-(a+b)\,] = 0 \quad (\text{as } f'(c) \neq 0)$$

hence $c = \frac{a+b}{2}$ also $c < \frac{a+b}{2}$, $\frac{dA}{dc} < 0$ and $c > \frac{a+b}{2}$, $\frac{dA}{dc} > 0$

Hence A is minimum when $c = \frac{a+b}{2}$

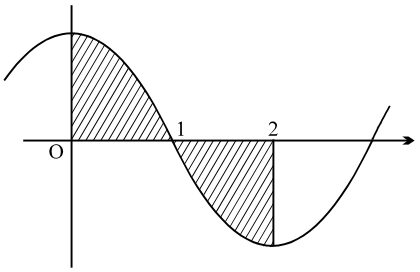
Questions

1. If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ and the straight lines $x = 0$; $x = 2$ and the x-axis is minimum then find the value of 'a'.

Ans. $\frac{2}{3}$

Sol. $f'(x) = x^2 - 2x = x(x-2) = 0$

(note that $f(x)$ is monotonic in $(0, 2)$)



Hence for the minimum and $f(x)$ must cross the x-axis are $\frac{0+2}{2}=1$

hence $f(1) = \frac{1}{3} - 1 + a = 0 \quad \Rightarrow \quad a = \frac{2}{3}$

2. Find the value of the parameter 'a' for which the area of the figure bounded by the abscissa axis, the graph of the function $y = x^3 + 3x^2 + x + a$, and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least.

Ans. $a = -1$

Sol. $f(x) = x^3 + 3x^2 + x + a$

$f'(x) = 3x^2 + 6x + 1 = 0 \qquad \Rightarrow \quad x = -1 \pm \frac{\sqrt{6}}{3}$

hence $f(x)$ cuts the x axis at $\frac{1}{2}\left[\left(-1+\frac{\sqrt{6}}{3}\right)+\left(-1-\frac{\sqrt{6}}{3}\right)\right] = -1$

$f(-1) = -1 + 3 - 1 + a = 0$

$a = -1$

3. If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is least. Find k and also the least area.

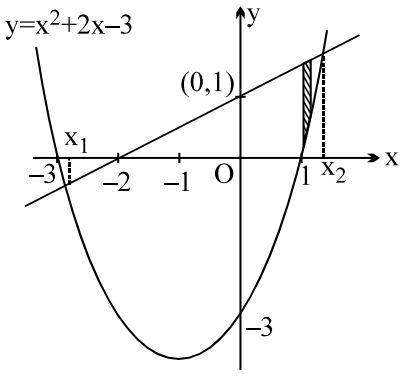
Ans. $k = 2, A_{min} = \frac{32}{3}$

Sol. x_1 and x_2 are the roots of the equation

$x^2 + 2x - 3 = kx + 1$

$x^2 + (2-k)x - 4 = 0$

$$\left.\begin{aligned} x_1 + x_2 &= k - 2 \\ x_1 x_2 &= -4 \end{aligned}\right\}$$



$$A = \int_{x_1}^{x_2} [(kx+1) - (x^2 + 2x - 3)]\,dx$$

$$= \left[(k-2)\frac{x^2}{2} - \frac{x^3}{3} + 4x\right]_{x_1}^{x_2} = \left[(k-2)\frac{x_2^2 - x_1^2}{2} - \frac{1}{3}\left(x_2^3 - x_1^3\right) + 4(x_2 - x_1)\right]$$

$$= (x_2 - x_1)\left[\frac{(k-2)^2}{2} - \frac{1}{3}\left((x_2 + x_1)^2 - x_1x_2\right) + 4\right]$$

$$= \sqrt{(x_2 + x_1)^2 - 4x_1x_2}\ \left[\frac{(k-2)^2}{2} - \frac{1}{3}\left((k-2)^2 + 4\right) + 4\right]$$

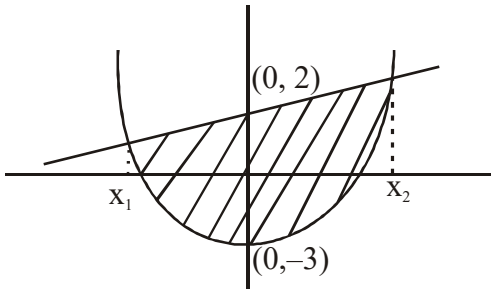
$$= \frac{\sqrt{(k-2)^2 + 16}}{6}\left[\frac{1}{6}(k-2)^2 + \frac{16}{3}\right] = \frac{[(k-2)^2 + 16]^{3/2}}{6}$$

$\Rightarrow$ A is minimum if $k = 2$. Hence $A_{min} = \frac{32}{3}$

4. For what value of k is the area of the figure bounded by the curves $y = x^2 - 3$ and $y = kx + 2$ is the least. Determine the least area.

Ans. $k = 0, A = \frac{20\sqrt{5}}{3}$

Sol.



$y = x^2 - 3$
$y = kx + 2$
$\Rightarrow x^2 - 3 = kx + 2$
$\Rightarrow x^2 - kx - 5 = 0$

$$\left.\begin{matrix} x_1 + x_2 = k \\ x_1x_2 \end{matrix}\right\} (x_2 - x_1) = \sqrt{k^2 + 20}$$

$$A = \int_{x_1}^{x_2} (kx + 2) - (x^2 - 3)\,dx = \int_{x_1}^{x_2} (k + 5 - x^2)\,dx$$

$$= \left(\frac{kx^2}{2} + 5x - \frac{x^3}{3}\right)_{x_1}^{x_2} = \left[\frac{k}{2}(x_2^2 - x_1^2) + 5(x_2 - x_1) - \frac{1}{3}(x_2^3 - x_1^3)\right]$$

$$= (x_2 - x_1)\left[\frac{k}{2}(x_2 + x_1) + 5 - \frac{1}{3}\left((x_2 + x_1)^2 - x_1x_2\right)\right]$$

$$= \sqrt{k^2+20}\left[\frac{k^2}{2}+5-\frac{1}{3}\left(k^2+5\right)\right]$$

$$A = \sqrt{k^2+20}\left[\frac{k^2}{6}+\frac{10}{3}\right]$$

$$A_{min} = \sqrt{20}\cdot\frac{10}{3} \qquad (k=0)$$

$$= \frac{20\sqrt{5}}{3}$$

6.1 AVERAGE VALUE OF A FUNCTION

Average value of a function $y = f(x)$ over an interval $a \le x \le b$ is defined as : $y(av) = \frac{1}{b-a}\int_a^b f(x)dx$.

Root mean square value **(RMS)** is defined as

$$\left[\rho = \frac{1}{b-a}\int_a^b f^2(x)\,dx\right]^{\frac{1}{2}}$$

Note :

(1) Average value can be + ve, – ve or zero.

(2) If the function is defined in $(0, \infty)$ then

$$y_{av} = \lim_{b\to\infty} \frac{1}{b}\int_0^b f(x)\,dx$$ provided the limit exists.

Questions

1. Compute the average value of $f(x) = \frac{\cos^2 x}{\sin^2 x + 4\cos^2 x}$ in $\left[0, \frac{\pi}{2}\right]$

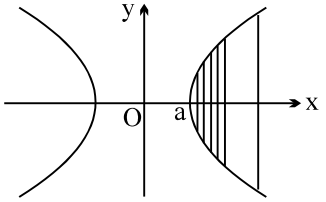
Ans. $\frac{1}{6}$

Sol. $$y_{av} = \frac{1}{\frac{\pi}{2}-0}\int_0^{\pi/2}\frac{\cos^2 x}{\sin^2 x + 4\cos^2 x}dx = \frac{2}{\pi}\int_0^{\pi/2}\frac{\sec^2 x}{(4+\tan^2 x)(1+\tan^2 x)}dx$$

$$= \frac{2}{\pi}\int_0^{\infty}\frac{dt}{(4+t^2)(1+t^2)} = \frac{2}{3\pi}\int_0^{\infty}\frac{(4+t^2)-(1+t^2)\,dt}{(4+t^2)(1+t^2)}$$

2. Find the average length of all vertical chords of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ over the interval $a \le x \le 2a$.

Ans. $b\left(2\sqrt{3}-\ell n\left(2+\sqrt{3}\right)\right)$



Sol. $y^2=\left(\frac{x^2}{a^2}-1\right)b^2 \quad \Rightarrow \quad y=\frac{b}{a}\sqrt{x^2-a^2}$

$$\frac{1}{2}y_{av.}=\frac{1}{2a-a}\int_a^{2a}\frac{b}{a}\sqrt{x^2-a^2}\,dx$$

$$y_{av.}=\frac{2b}{a^2}\int_a^{2a}\sqrt{x^2-a^2}\,dx=\frac{2b}{a^2}\left[\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\ell n\left(x+\sqrt{x^2+a^2}\right)\right]_a^{2a}$$

3. Find the average value of y^2 w.r.t. x for the curve $ay=b\sqrt{a^2-x^2}$ between $x=0$ & $x=a$. Also find the average value of y w.r.t. x^2 for $0 \le x \le a$.

Ans. (i) $f(x)\big|_{av}=\frac{2b^2}{3}$, (ii) $f(x)\big|_{av}=\frac{2b}{3}$

Sol. Let $f(x)=y^2=\frac{b^2}{a^2}(a^2-x^2)$.

Now $f(x)\big|_{av}=\frac{b^2}{a^2(a-0)}\int_0^a(a^2-x^2)\,dx=\frac{2b^2}{3}$

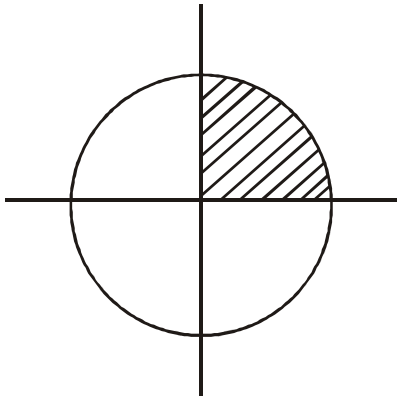
Again y_{av} w.r.t. x^2 is

$$f(x)\big|_{av}=\frac{1}{(a^2-0)}\int_0^{a^2}y\,d(x^2)=\frac{b}{a^2a}\int_0^{a^2}\sqrt{a^2-x^2}\,dx^2=\frac{b}{a^3}\int_0^{a^2}\sqrt{a^2-t}\,dt=\frac{2b}{3}$$

6.2 AREA WHEN CURVE IS GIVEN IN PARAMETRIC FORM

Q. Find the area made by $x=2\cos\theta$ and $y=2\sin\theta$.

Ans. 4π

Sol. 

$x = 2\cos\theta \qquad y = 2\sin\theta$

$$\text{Area} = \int_0^2 y.dx \quad = \int_{\frac{\pi}{2}}^{\theta} (2\sin\theta)(-2\sin\theta)\,d\theta = 4\int_0^{\frac{\pi}{2}} \sin^2\theta.d\theta$$

6. DETERMINATION OF FUNCTION

1. The area from 0 to x under a certain graph is given to be $A = \sqrt{1+3x} - 1, x \geq 0$;

(a) Find the average rate of change of A w.r.t. x as x increases from 1 to 8.
(b) Find the instantaneous rate of change of A w.r.t. x at x = 5.
(c) Find the ordinate (height) y of the graph as a function of x.
(d) Find the average value of the ordinate (height) y, w.r.t. x as x increases from 1 to 8

Ans: **(a)** $\frac{3}{7}$ **(b)** $\frac{3}{8}$ **(c)** $y = \frac{3}{2\sqrt{1+3x}}$ **(d)** $\frac{3}{7}$]

Sol.

(a) $A(1) = 1 \qquad A(8) = 4$

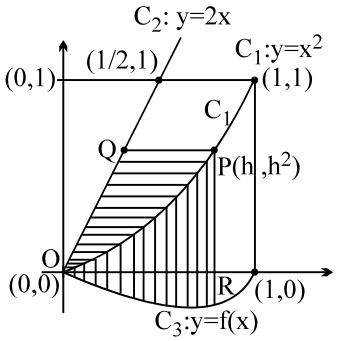
$$\frac{A(8) - A(1)}{8-1} = \frac{3}{7} \text{ Ans.}$$

(b) $$\left.\frac{dA}{dx}\right|_{x=5} = \frac{1\cdot 3}{2\sqrt{1+3x}} = \frac{3}{8} \text{ Ans.}$$

(c) $y = \frac{3}{2\sqrt{1+3x}}$ Ans.

(d) $$\frac{1}{(8-1)}\int_1^8 \frac{3}{2\sqrt{1+3x}}dx = \frac{1}{7}\int_1^8 \frac{3}{2\sqrt{1+3x}}dx = \frac{3}{7} \text{ Ans.]}$$

2. Let C_1 & C_2 be the graphs of the functions $y = x^2$ & $y = 2x$, $0 \leq x \leq 1$ respectively. Let C_3 be the graph of a function $y = f(x)$, $0 \leq x \leq 1$, $f(0) = 0$. For a point P on C_1, let the lines through P, parallel to the axes, meet C_2 & C_3 at Q & R respectively (see figure). If for every position of P (on C_1), the areas of the shaded regions OPQ & ORP are equal, determine the function f(x). [JEE '98, 8]



Ans. $f(x) = x^3 - x^2$

Sol. $\int_0^{h^2} \left(\sqrt{y} - \frac{y}{2} \right) dy = \int_0^{h} \left(x^2 - f(x) \right) dx$

differentiate both sides w.r.t. h

$$\left(h - \frac{h^2}{2} \right) 2h = h^2 - f(h)$$

$$f(h) = h^2 - \left(h - \frac{h^2}{2} \right) 2h = h^2 - h(2h - h^2) = h^2 - 2h^2 + h^3$$

$$= h^3 - h^2$$

$$f(x) = x^3 - x^2$$

$$= x^2(x - 1)$$